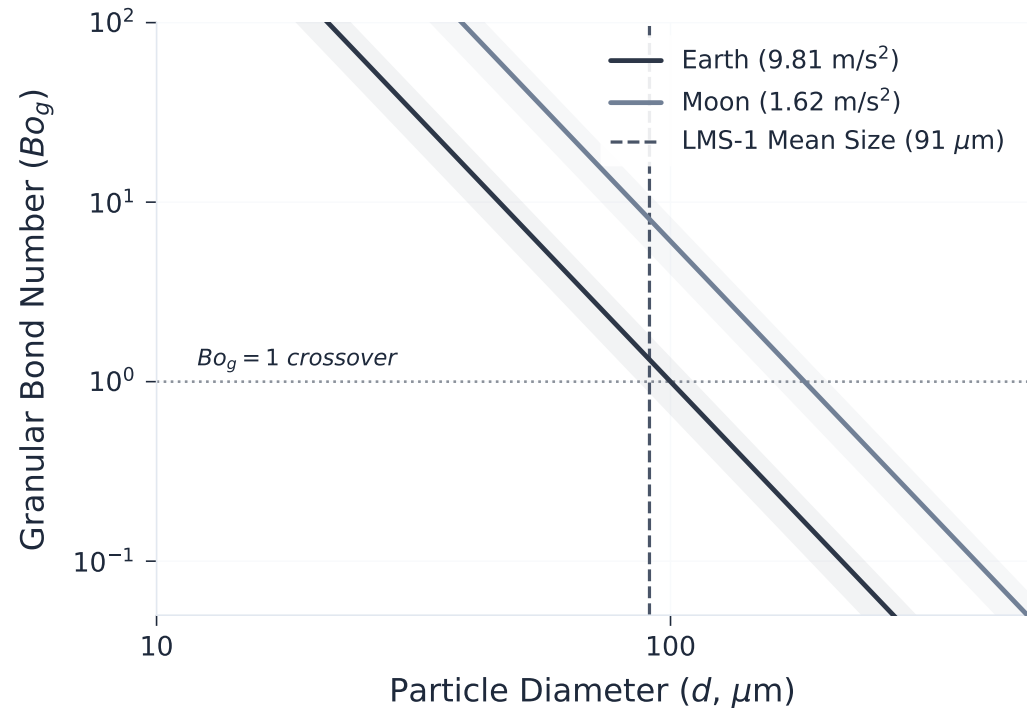


(a) Analytical Granular Bond Number**(b) Charge Accumulation History**